

# Patent Disclosure and Reliance on External Financing

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LEHIGH  
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- ⇒ Firms hide knowledge by writing opaque patents (“**obfuscation**”)

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- Substantial support for innovation  $\leftrightarrow$  knowledge disclosure
- A crack in the bargain: clear disclosure has real costs
  - Accelerates imitation
  - Increases infringement due to clearer patent scope $\Rightarrow$  Firms hide knowledge by writing opaque patents (“**obfuscation**”)
- We ask:
  1. Can we measure obfuscation? (**Yes.**)
  2. What disciplines the incentive to obfuscate? $\Rightarrow$  **Reliance on external financing plays a central role**

## The paper in a slide

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- **Key innovation:** LLM counterfactual decomposes patent complexity
  - $\text{Obfuscation} = \text{Total Complexity} - \text{Fundamental Complexity (LLM rewrite)}$   
Validation:  $\uparrow$  in information asymmetry,  $\downarrow$  in trade secrets

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⇒ **A new channel for financing to shape the trajectory of innovation**

## We decompose complexity into fundamental and strategic components

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- Problem: Some technologies are genuinely harder to describe
- Solution: Use an LLM (Sonnet 3.5) to construct a counterfactual abstract  
*Give me a simpler and clearer version of this patent abstract. Reply with JUST your version, not any label, title, header, or comments.*
- Scale: 2.6M patents matched to public firms, 1976–2023
- Measurement: For any linguistic complexity measure  $f(\cdot)$ :

$$\begin{aligned}\text{Obfuscation} &= f(\text{Original abstract}) - f(\text{LLM rewrite}) \\ &= \text{Total Complexity} - \text{Fundamental Complexity}\end{aligned}$$

Appendix: [Meaning preservation] and [LLM output stability]

# IBM Patent 9547842: 7 grade levels of additional complexity

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## As filed:

*A method, a system and a computer program product of transmitting availability information in an electronic mail system are disclosed. The system includes a first system including a sender, the first system is adapted to activate an availability indicator having a start date, wherein the activation is performed by the sender and extract availability indicator metadata from the availability indicator. ...*

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## LLM rewrite:

This invention describes a way to share availability information through email. It involves a system where a sender can set up an availability indicator with start and end dates, along with a message. When the sender emails someone for the first time after setting this up, and before the start date, the system automatically adds the availability information to the email. The system also includes a scheduling feature that helps create these availability indicators.

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## Gunning-Fog:

12.81 (college freshman)

# Descriptive Facts

## Validation: Information asymmetry is higher following obfuscated grants

|                                 | Info Asymmetry Percentile |                    |                    | Trade Secret (Boolx100) |                    |                    |
|---------------------------------|---------------------------|--------------------|--------------------|-------------------------|--------------------|--------------------|
|                                 | (1)                       | (2)                | (3)                | (4)                     | (5)                | (6)                |
| Obfuscation (GF) (z)            | 0.423***<br>(0.09)        | 0.658***<br>(0.17) | 0.410***<br>(0.09) | -0.338**<br>(0.15)      | -0.671**<br>(0.33) | -0.320**<br>(0.16) |
| Total Complexity (GF) (z)       |                           | -0.245<br>(0.18)   |                    |                         | 0.348<br>(0.36)    |                    |
| Fundamental Complexity (GF) (z) |                           |                    | -0.060<br>(0.04)   |                         |                    | 0.085<br>(0.09)    |
| Fixed effects:                  |                           |                    |                    |                         |                    |                    |
| CPC-year                        | Yes                       | Yes                | Yes                | Yes                     | Yes                | Yes                |
| Obs.                            | 1,802,286                 | 1,802,286          | 1,802,286          | 1,455,104               | 1,455,104          | 1,455,104          |
| $R^2$                           | 0.49                      | 0.49               | 0.49               | 0.27                    | 0.27               | 0.27               |

- Validation a la Bushee et al. (2018), using Amihud (2002) illiquidity
- Obfuscation is stronger signal than total patent complexity

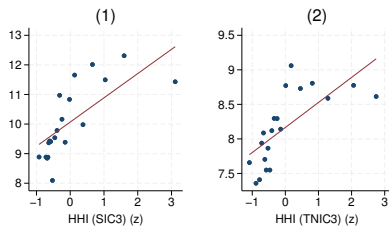
# Obfuscation is a form of IP protection

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- 1 s.d.  $\rightarrow$  0.34p.p. less likely to discuss trade secrets (Glaeser (2018)) in 10-K
- Suggests that obfuscation substitutes for trade secrets

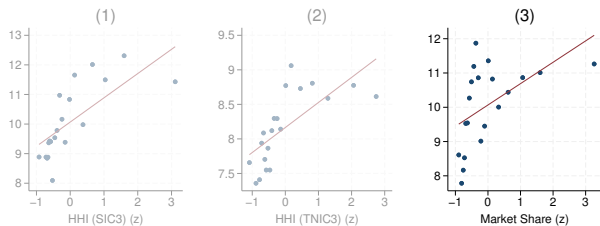
# Competitive factors as determinants of obfuscation

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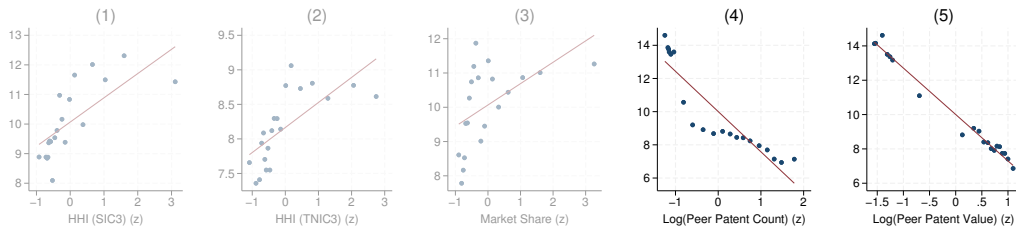
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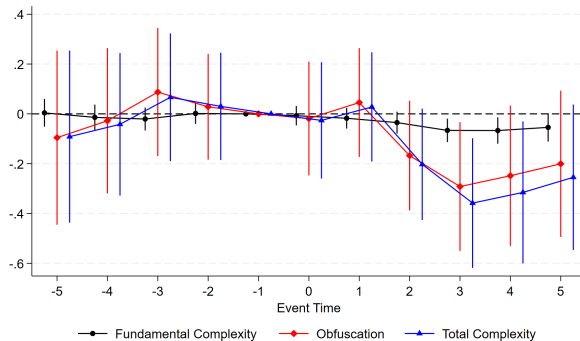
- Rises when facing threats from dominant industry leaders
- Rises when the firm has a larger market share to defend

# Competitive factors as determinants of obfuscation



- Rises when facing threats from dominant industry leaders
- Rises when the firm has a larger market share to defend
- Falls after competitors develop more valuable patents

# When firms pledge patents as collateral, obfuscation falls



- Event study around a firm's **first patent pledge**
- Pledging makes financing value of patents salient
- Result: complexity falls, driven by obfuscation decline

First evidence of a **financial factor** disciplining the obfuscation decision

# Four sources of identification

## Four strategies to identify the strategic-financing tradeoff

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- **Strategy 1: Examiner IV** random assignment of patent examiners
  - Some examiners enforce greater §112 clarity (Dyer et al. 2024)
- **Strategy 2: Nautilus v. Biosig (2014)** Supreme Court clarity mandate
  - Articulated a stronger uniform requirement for definiteness of claims (Ashtor 2022)
- **Strategy 3: AIPA (1999)** accelerated disclosure requirement
  - Patents filed after 2000 published 18 months after filing unless US-only (Ireland et al. 2024)
- **Strategy 4: Moody's pension adjustment (2006)** shock to financial constraints
  - Reduced adjusted leverage, positively shocked ratings (Kraft 2015; Kisgen 2019)

## IV tests demonstrate strategic/financing tradeoff

|                          | 1st<br>Stg          | Log of 1 plus cites received from |                     |                     | Other outcomes      |                     |                    |
|--------------------------|---------------------|-----------------------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
|                          | (1)<br>Obf. (z)     | (2)<br>All                        | (3)<br>Others       | (4)<br>Self         | (5)<br>Infringed    | (6)<br>PatSold      | (7)<br>Pledged     |
| Ex. 112 Strictness (z)   | -0.010***<br>(0.00) |                                   |                     |                     |                     |                     |                    |
| <i>Obfuscation(z)</i>    |                     | -2.162***<br>(0.47)               | -1.273***<br>(0.32) | -1.515***<br>(0.33) | -0.046***<br>(0.02) | -0.381***<br>(0.12) | -0.202**<br>(0.08) |
| Ex. Other Strictness (z) | -0.009***<br>(0.00) | -0.006<br>(0.00)                  | -0.005*<br>(0.00)   | -0.002<br>(0.00)    | -0.000<br>(0.00)    | -0.001<br>(0.00)    | 0.001<br>(0.00)    |
| Fund. Complexity (z)     | -0.200***<br>(0.00) | -0.409***<br>(0.09)               | -0.236***<br>(0.06) | -0.292***<br>(0.07) | -0.009***<br>(0.00) | -0.074***<br>(0.02) | -0.039**<br>(0.02) |
| Fixed effects:           |                     |                                   |                     |                     |                     |                     |                    |
| CPC-year                 | Y                   | Y                                 | Y                   | Y                   | Y                   | Y                   | Y                  |
| Art-unit-year            | Y                   | Y                                 | Y                   | Y                   | Y                   | Y                   | Y                  |
| Obs.                     | 733,013             | 733,013                           | 733,013             | 733,013             | 733,013             | 733,013             | 733,013            |
| Cragg-Donald F           | 27.71               |                                   |                     |                     |                     |                     |                    |

**IV relevance:** §112-strict examiners reduce patent obfuscation (F-stat=27.7)

*In paper: Direct evidence that strict examiners lead to simplified language during review*

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**Benefits:** External follow-on innovation and infringement declines  
*Within-firm* decline consistent with reduced need for thickets

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**(Financial) Costs:** Reduces patent sales (value and liquidity) and pledging

# Financing-dependent firms respond less to Nautilus clarity mandate

| Z variable:         | (1)<br>None         | (2)<br>Pledged      | (3)<br>HighLev      | (4)<br>None        | (5)<br>Pledged      | (6)<br>HighLev      |
|---------------------|---------------------|---------------------|---------------------|--------------------|---------------------|---------------------|
| Exposure            | 1.214***<br>(0.03)  | 1.072***<br>(0.06)  | 1.242***<br>(0.03)  |                    |                     |                     |
| Post                | 1.478***<br>(0.33)  | 3.668***<br>(0.72)  | 2.109***<br>(0.46)  | 0.420**<br>(0.21)  | 1.488***<br>(0.45)  | 0.830***<br>(0.19)  |
| Exposure x Post     | -0.158***<br>(0.05) | -0.512***<br>(0.12) | -0.271***<br>(0.08) | -0.071**<br>(0.03) | -0.254***<br>(0.07) | -0.138***<br>(0.03) |
| Z                   |                     | -0.226<br>(0.17)    | 0.967***<br>(0.36)  |                    |                     | -0.509<br>(0.41)    |
| Z x Exposure        |                     | 0.061**<br>(0.03)   | -0.081<br>(0.05)    |                    |                     |                     |
| Z x Post            |                     | -1.016***<br>(0.28) | -1.767***<br>(0.62) |                    | -0.455**<br>(0.18)  | -1.105***<br>(0.36) |
| Z x Exposure x Post |                     | 0.163***<br>(0.05)  | 0.292***<br>(0.10)  |                    | 0.078***<br>(0.03)  | 0.174***<br>(0.06)  |
| Fixed effects:      |                     |                     |                     |                    |                     |                     |
| Year                | N                   | N                   | N                   | Y                  | Y                   | Y                   |
| Firm                | N                   | N                   | N                   | Y                  | Y                   | Y                   |
| Obs.                | 503,093             | 503,093             | 503,086             | 503,093            | 503,093             | 503,086             |
| R <sup>2</sup>      | 0.12                | 0.12                | 0.12                | 0.14               | 0.14                | 0.14                |

- *Nautilus* (2014) raises the requirement for claim clarity
  - Ashtor (2022): §112 rejections ↑
  - We use DD/DDD design
  - Exposure: Obfu > p50 pre-event

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- *Nautilus* reduced obfuscation
- But less so for firms that
  - already use patents as collateral

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| Z variable:         | (1)<br>None         | (2)<br>Pledged      | (3)<br>HighLev      | (4)<br>None        | (5)<br>Pledged      | (6)<br>HighLev      |
|---------------------|---------------------|---------------------|---------------------|--------------------|---------------------|---------------------|
| Exposure            | 1.214***<br>(0.03)  | 1.072***<br>(0.06)  | 1.242***<br>(0.03)  |                    |                     |                     |
| Post                | 1.478***<br>(0.33)  | 3.668***<br>(0.72)  | 2.109***<br>(0.46)  | 0.420**<br>(0.21)  | 1.488***<br>(0.45)  | 0.830***<br>(0.19)  |
| Exposure x Post     | -0.158***<br>(0.05) | -0.512***<br>(0.12) | -0.271***<br>(0.08) | -0.071**<br>(0.03) | -0.254***<br>(0.07) | -0.138***<br>(0.03) |
| Z                   |                     | -0.226<br>(0.17)    | 0.967***<br>(0.36)  |                    |                     | -0.509<br>(0.41)    |
| Z x Exposure        |                     | 0.061**<br>(0.03)   | -0.081<br>(0.05)    |                    |                     |                     |
| Z x Post            |                     | -1.016***<br>(0.28) | -1.767***<br>(0.62) |                    | -0.455**<br>(0.18)  | -1.105***<br>(0.36) |
| Z x Exposure x Post |                     | 0.163***<br>(0.05)  | 0.292***<br>(0.10)  |                    | 0.078***<br>(0.03)  | 0.174***<br>(0.06)  |
| Fixed effects:      |                     |                     |                     |                    |                     |                     |
| Year                | N                   | N                   | N                   | Y                  | Y                   | Y                   |
| Firm                | N                   | N                   | N                   | Y                  | Y                   | Y                   |
| Obs.                | 503,093             | 503,093             | 503,086             | 503,093            | 503,093             | 503,086             |
| R <sup>2</sup>      | 0.12                | 0.12                | 0.12                | 0.14               | 0.14                | 0.14                |

- *Nautilus* (2014) raises the requirement for claim clarity
  - Ashtor (2022): §112 rejections ↑
  - We use DD/DDD design
  - Exposure: Obfu > p50 pre-event
- *Nautilus* reduced obfuscation
- But less so for firms that
  - already use patents as collateral
  - rely more on external financing

→ Finance-dependent firms already emphasized transparency and had less room to respond

# Financial dependence dampens the AIPA obfuscation response

| Z variable:         | (1)<br>None         | (2)<br>Pledged       | (3)<br>HighLev       | (4)<br>None         | (5)<br>Pledged   | (6)<br>HighLev      |
|---------------------|---------------------|----------------------|----------------------|---------------------|------------------|---------------------|
| Exposure            | 5.312***<br>(0.381) | 1.220<br>(0.825)     | 4.260***<br>(0.483)  |                     |                  |                     |
| Post                | -0.541*<br>(0.311)  | -1.361***<br>(0.439) | -0.801**<br>(0.374)  |                     |                  |                     |
| Exposure x Post     | 1.266**<br>(0.543)  | 4.200***<br>(1.168)  | 2.904***<br>(0.700)  | 1.192***<br>(0.238) | 0.755<br>(0.579) | 1.621***<br>(0.281) |
| Z                   |                     | -0.856*<br>(0.449)   | 0.479<br>(0.481)     |                     |                  | -0.607*<br>(0.313)  |
| Z x Exposure        |                     | 4.497***<br>(0.919)  | 1.753***<br>(0.648)  |                     |                  | 0.750*<br>(0.416)   |
| Z x Post            |                     | 0.861<br>(0.553)     | 1.045*<br>(0.597)    |                     | 0.529<br>(0.386) | 0.740**<br>(0.325)  |
| Z x Exposure x Post |                     | -2.930**<br>(1.308)  | -3.842***<br>(0.963) |                     | 0.478<br>(0.629) | -1.111**<br>(0.441) |
| Fixed effects:      |                     |                      |                      |                     |                  |                     |
| Year                | N                   | N                    | N                    | Y                   | Y                | Y                   |
| Firm                | N                   | N                    | N                    | Y                   | Y                | Y                   |
| Obs.                | 982,497             | 982,497              | 976,986              | 982,496             | 982,496          | 976,985             |
| R <sup>2</sup>      | 0.02                | 0.02                 | 0.02                 | 0.10                | 0.10             | 0.10                |

- AIPA speeds up disclosure 20 months for patents also filed abroad
  - Exposure: % patents co-filed abroad
  - Ireland et al. (2024): → Total Complexity

# Financial dependence dampens the AIPA obfuscation response

| Z variable:         | (1)<br>None         | (2)<br>Pledged       | (3)<br>HighLev       | (4)<br>None         | (5)<br>Pledged   | (6)<br>HighLev      |
|---------------------|---------------------|----------------------|----------------------|---------------------|------------------|---------------------|
| Exposure            | 5.312***<br>(0.381) | 1.220<br>(0.825)     | 4.260***<br>(0.483)  |                     |                  |                     |
| Post                | -0.541*<br>(0.311)  | -1.361***<br>(0.439) | -0.801**<br>(0.374)  |                     |                  |                     |
| Exposure x Post     | 1.266**<br>(0.543)  | 4.200***<br>(1.168)  | 2.904***<br>(0.700)  | 1.192***<br>(0.238) | 0.755<br>(0.579) | 1.621***<br>(0.281) |
| Z                   |                     | -0.856*<br>(0.449)   | 0.479<br>(0.481)     |                     |                  | -0.607*<br>(0.313)  |
| Z x Exposure        |                     | 4.497***<br>(0.919)  | 1.753***<br>(0.648)  |                     |                  | 0.750*<br>(0.416)   |
| Z x Post            |                     | 0.861<br>(0.553)     | 1.045*<br>(0.597)    |                     | 0.529<br>(0.386) | 0.740**<br>(0.325)  |
| Z x Exposure x Post |                     | -2.930**<br>(1.308)  | -3.842***<br>(0.963) |                     | 0.478<br>(0.629) | -1.111**<br>(0.441) |
| Fixed effects:      |                     |                      |                      |                     |                  |                     |
| Year                | N                   | N                    | N                    | Y                   | Y                | Y                   |
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| Z                   |                     | -0.856*<br>(0.449)   | 0.479<br>(0.481)     |                     |                  | -0.607*<br>(0.313)  |
| Z x Exposure        |                     | 4.497***<br>(0.919)  | 1.753***<br>(0.648)  |                     |                  | 0.750*<br>(0.416)   |
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→ The cost of increasing obfuscation is higher for finance-dependent firms

## Easing financial constraints unlocks more obfuscation

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- Shock to **financial constraints**: Moody's 2006 pension adjustment
- ↑ credit-rating of firms with large underfunded pensions (Kraft 2015; Kisgen 2019)

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|                    | (1)               | (2)                    |
|--------------------|-------------------|------------------------|
|                    | Obfuscation       | Fundamental Complexity |
| Underfunded × Post | 0.584**<br>(0.25) | -0.043<br>(0.04)       |
| Fixed effects:     |                   |                        |
| CPC-year           | Y                 | Y                      |
| Firm               | Y                 | Y                      |
| Obs.               | 151,618           | 151,618                |
| $R^2$              | 0.05              | 0.08                   |

## Summary of tests

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- **A firm's first pledge** — salience of patents' financing role increases  
→ Obfuscation of subsequent patents falls

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**Firms balance opaque disclosure in patents against their need to attract external capital**

- **A new measurement approach:** LLM counterfactuals decompose strategic from fundamental complexity
  - Applicable in many disclosure settings
  - Inspired Bushee et al. (2018) decomp
  - Bai et al. (2023): *What* is disclosed?
  - This paper: *How* (clearly) is it disclosed?

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**Financial considerations discipline patent obfuscation and innovation disclosure — a new channel through which finance can shape the trajectory of innovation.**

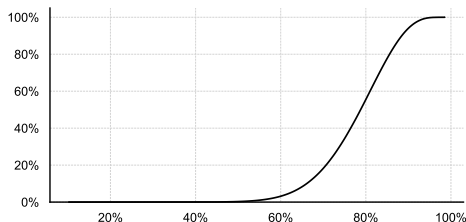
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# Appendix:

# Method validation: semantic fidelity and cross-LLM robustness

## LLMs preserve meaning

- *PatentSBERTa* embeddings (Bekamiri et al., 2024)
- Cosine Similarity(original, rewritten) abstracts
- Empirical CDF of Cosine Similarity:



## Stability across LLMs and runs

- Unreported: Repeated-run correlations  $\geq 99\%$
- Correlations between LLMs  $\geq 92\%$ :

| Model                              | Legend | (1)  | (2)  | (3)  | (4)  | (5)  | (6)  | (7) |
|------------------------------------|--------|------|------|------|------|------|------|-----|
| anthropic/claude-3-5               | (1)    |      |      |      |      |      |      |     |
| anthropic/claude-3-7-sonnet-latest | (2)    | 0.97 |      |      |      |      |      |     |
| anthropic/claude-4-opus            | (3)    | 0.96 | 0.94 |      |      |      |      |     |
| google/gemini-2.5-flash-lite       | (4)    | 0.94 | 0.95 | 0.91 |      |      |      |     |
| meta/Llama-3.3-70B-Instruct-Turbo  | (5)    | 0.92 | 0.93 | 0.90 | 0.91 |      |      |     |
| openai/gpt-4o                      | (6)    | 0.98 | 0.98 | 0.94 | 0.95 | 0.93 |      |     |
| xai/grok-4-1-fast-non-reasoning    | (7)    | 0.96 | 0.97 | 0.93 | 0.94 | 0.92 | 0.97 |     |

← Back to measurement slide